Elliptic Curve Cryptography-1

Module III

Introduction

- An elliptic curve is the set of points that satisfy a specific mathematical equation.

$$y^2 = x^3 + ax + b (for\ some\ constants)$$

$$Determinant\ 4a^3 + 27b^2 \neq 0$$

E.g. (x,y)=(2,4) for a=3, b=2.

$$y^2 = x^3 + ax + b$$

$$4^2 = 2^3 + 3.2 + 2$$

$$16 = 8 + 6 + 2$$

$$16 = 16 \checkmark$$

$$4a^3 + 27b^2 \neq 0$$

$$4.3^3 + 27.2^2 \neq 0$$

$$4.27 + 27.4 \neq 0 \checkmark$$

Types of Elliptic curves

1. Elliptic curves on real numbers.
2. Elliptic curves on complex numbers.
3. Elliptic curves on finite fields. E.g. GF(2), GF(5), GF(8).

Hence the equation for elliptic curve over finite field,

$$y^2 \equiv x^3 + ax + b \ (mod\ p), a < p, b < p$$

$$4a^3 + 27b^2 \ (mod\ p) \neq 0$$

How to construct an elliptic curve over finite field?

- Consider *Z/11 = {0 , 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}.*
- p=11, a=1, b=1. ➡ $4a^3+27b^2\ (mod\ p) \neq 0$

$$4.1^3+27.1^2\ (mod\ 11) \neq 0$$

$$4+27\ (mod\ 11) \neq 0$$

$$31\ (mod\ 11) \neq 0$$

$$31\ (mod\ 11) \neq 0\ \checkmark$$

How to construct an elliptic curve over finite field? Cntd..

$$y^2 \equiv x^3 + ax + b \ (mod\ p), a < p, b < p$$

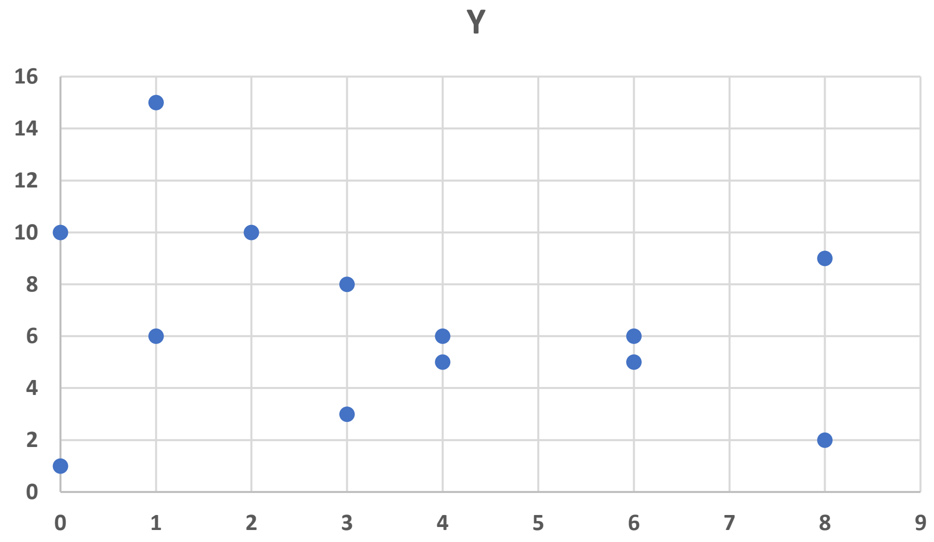
$$y^2 mod\ p = x^3 + ax + b \ (mod\ p)$$

$$y^2 mod\ 11 = x^3 + x + 1 \ (mod\ 11)$$

Set of points generated:

(0,1), (0,10), (1,15), (1,6), (2,10), (3,3), (3,8), (4,5), (4,6), (6,5), (6,6), (8,2), (8,9).

Plot



	$y^2 mod\ 11$ **(mod p)**	$x^3 + x + 1$ **(mod p)**
0	0 (y=0)	1 (x=0)
1	1 (y=1)	3 (x=1)
2	4 (y=2)	0 (x=2)
3	9 (y=3)	9 (x=3)
4	5 (y=4)	3 (x=4)
5	3 (y=5)	10 (x=5)
6	3 (y=6)	3 (x=6)
7	5 (y=7)	10 (x=7)
8	9 (y=8)	4 (x=8)
9	4 (y=9)	2 (x=9)
10	1 (y=10)	10 (x=10)

Checking if a point falls on the curve?

- (3,8)= $8^2 mod\ 11 = 3^3 + 3 + 1\ (mod\ 11)$

  $9 = 31\ (mod\ 11)$

  $9 = 9$ ✓